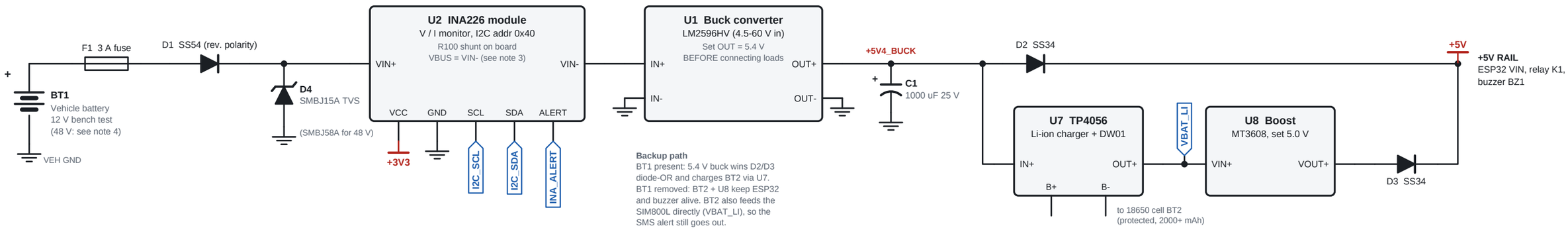
# EV VEHICLE TRACKER - PROTOTYPE SCHEMATIC (SIM800L / 2G)

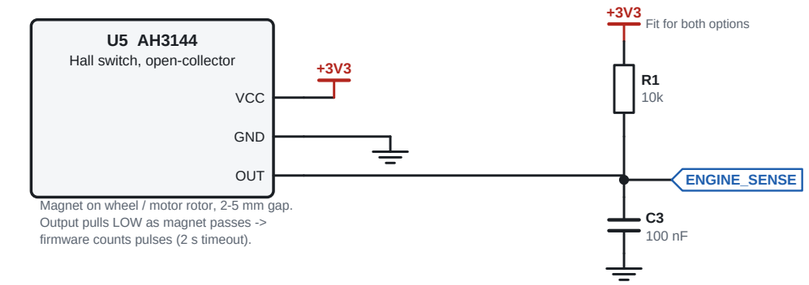
Matches firmware EV_Tracker_v3_SIM800L.ino (pin map frozen). Net labels with the same name are connected.

## A. POWER, PROTECTION, BACKUP & BATTERY SENSE

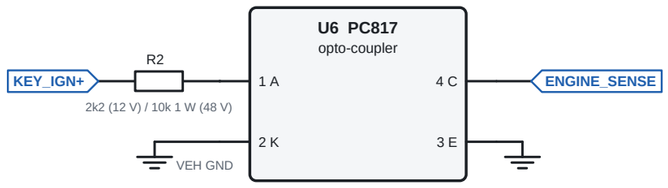



## B. ENGINE SENSE (fit ONE option)

**OPTION A - Hall sensor (ENGINE_SENSE_MODE 0)**

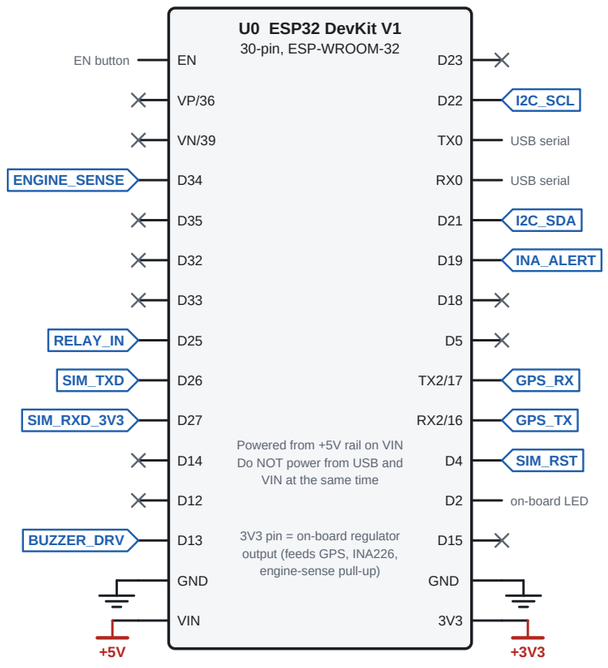



**OPTION B - Ignition sense (ENGINE_SENSE_MODE 1)**

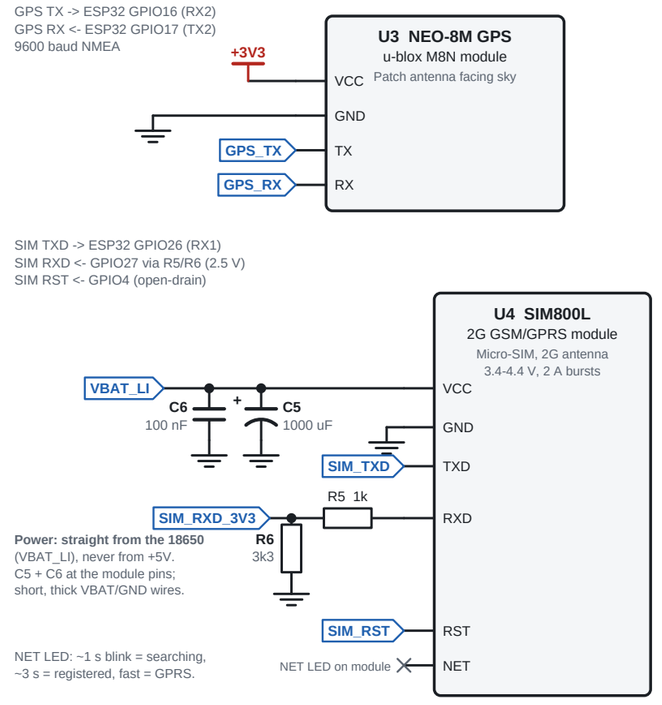



Ignition ON -> LED on -> transistor pulls ENGINE_SENSE LOW.
Isolates vehicle wiring from the ESP32.

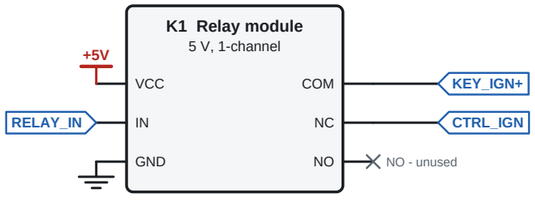
## C. MICROCONTROLLER




## D. GPS + 2G GSM MODULE (SIM800L)

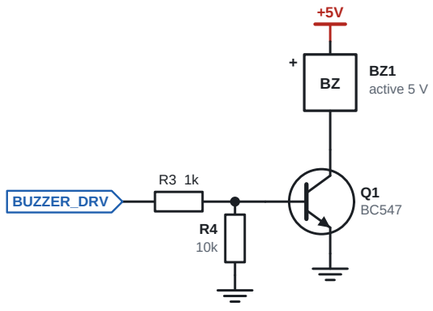



## E. IMMOBILISER RELAY (fail-safe)




KEY_IGN+ = key-switch output. CTRL_IGN = motor-controller ignition / key input.
Break the key line and pass it through COM-NC. Low-current signal only, never the traction line.
De-energised (boot, power loss, fault) -> NC closed -> vehicle RUNS.
Energised (app sends "OFF", speed <= 5 km/h) -> NC opens -> ignition CUT.
Set RELAY_ACTIVE_LOW in firmware to match your module.

## F. ALARM BUZZER




GPIO13 HIGH -> Q1 on -> buzzer sounds.
R4 holds Q1 off during boot.

## G. NOTES

1. All GND symbols are one common ground: vehicle battery -, buck, TP4056, boost, ESP32, modules.
2. GPIO34 is input-only with no internal pull-up: R1 (10k to +3V3) is mandatory.
3. INA226: power VCC from +3V3 (keeps I2C pull-ups at 3.3 V). VBUS max 36 V. On most boards VBUS is tied to VIN- on the PCB; if yours has a separate VBUS pin, connect it to VIN-.
4. 48 V vehicles: U1 must be a 60 V+ part (LM2596HV / XL7015), D4 = SMBJ58A, and the INA226 VBUS must see < 36 V (divider into VBUS + set BATT_DIVIDER_RATIO) or use an INA228 (85 V).
5. Build order: power -> backup (BT2) -> ESP32 -> SIM800L -> GPS -> INA226 -> engine sense -> buzzer -> relay. SIM800L works only on 2G (Airtel / Vi / BSNL SIM; Jio will NOT work).
6. Pins kept free of ESP32 boot-strapping pins (0, 2*, 5, 12, 15). *GPIO2 only drives on-board LED.
7. SIM800L logic is 2.8 V: R5/R6 drop the ESP32 TX to ~2.5 V. Never drive RST HIGH from the ESP32.

| EV Vehicle Tracker - SIM800L Prototype | |
|---|---|
| Doc: EVT-SCH-03 Rev 3.0 Sheet 1 of 1 | Date: 25 Oct 2026 |
| Firmware: EV_Tracker_v3_SIM800L.ino Board: ESP32 DevKit V1 | Size A3 |